Fiscal Year:	FY 2017	**Task Last Updated:**	FY 12/02/2016
PI Name:	Yu, Nan Ph.D.		
Project Title:	JPL Participation in ESA ACES (Atomic Clock Ensemble in Space) Worldwide Clock Comparison Campaign		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
Start Date:	10/01/2010	**End Date:**	03/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 3/31/2018 per PI (Ed., 2/19/15)		
Key Personnel Changes/Previous PI:	In addition to PI/Co-I, the following people have made contributions to the task in FY15: William Diener, Lin Yi, and Slava Turyshev.		
COI Name (Institution):	Burt, Eric (Jet Propulsion Laboratory) Tjoelker, Robert (Jet Propulsion Laboratory)		
Grant/Contract No.:	JPL Task Plan number 71-15743		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tremendous progress has been made recently in the U.S. and other countries in advancing atomic clock technology and its applications for the benefit of science, national interests, and society in general. Space provides a unique environment in which many benefits can be realized. The European Space Agency (ESA) is readying the ACES (Atomic Clock Ensemble in Space) payload for the International Space Station (ISS) for an anticipated 2013 launch. Some of the key objectives of ACES will be achieved by global time transfer and clock comparisons with the most advanced clocks at multiple worldwide sites. There is an opportunity for investigators from the leading U.S. clock and metrology labs to participate in the clock comparison campaign, enhancing the overall science objective of ACES and at the same time demonstrating and providing science benefit from advanced U.S. clock technology. The Jet Propulsion Laboratory (JPL) is one of the major U.S. time and frequency metrology institutions serving NASA missions and Deep Space Network needs. It was the NASA center for the microgravity LCAP program (Laser Cooling and Atomic Physics) in charge of developing the cold atom clock payload for PARCS (Primary Atomic Reference Clock in Space). JPL has also developed a unique atomic frequency standard that has demonstrated an exceptional long-term stability. We propose to collaborate with ESA on the ACES mission by participating in the ACES global clock campaign with JPL's ultra-stable mercury ion clock. During a proposed six-year performance period, we will: 1) contribute significantly to the ACES mission science objectives by providing unique clock references from a Western U.S. site; 2) provide and demonstrate a high-impact application for JPL's high performance time keeping clock technology; and 3) re-establish NASA's involvement in ISS Clock Science and initiate a healthy collaboration between NASA and ESA in the area of clock and quantum sensors. The task is to collaborate with ESA on the ACES mission by participating in the ACES global clock comparison campaign and provide frequency and time references derived from JPL's ultra-stable mercury ion clock. These references will be connected to a microwave link ground terminal provided by ACES and installed at JPL. Clock comparisons will be carried out over the microwave link with ESA clocks on the ISS. As JPL is the only ground comparison site located in the western USA and the mercury ion clock is the only one of its kind in the global clock comparison campaign, JPL's participation will contribute significantly to the overall ACES mission science objectives, while simultaneously providing a high-impact application for JPL's high performance time keeping clock technology. The task will also help re-establish NASA's involvement in ISS clock science and enable future collaboration between NASA and ESA in the area of clocks and quantum sensors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Advanced atomic clocks are critical to the interests of NASA and the nation. Telecommunication and navigation (e.g., GPS) are well-known applications of space clocks. Within NASA, high performance atomic clocks and frequency and timing capabilities enable operation of the NASA Deep Space Network and provide fundamental tools for precise measurements exploring the laws of Nature. Progress in advanced clock technology development, both on ground and for space, is rapidly being made in many countries. Participation by the U.S. in the ACES mission contributes to the science objectives of the ISS consistent with the 2011 National Research Council (NRC) report recommendations in Fundamental physics and demonstrates the capability and potential of NASA mercury ion clock technology.
Task Progress:	Before the ACES clock payload is launched to ISS, the JPL's effort has been in three main areas: upgrading the JPL ground clocks, preparation and installation of the microwave link ground terminal at JPL site, and investigation of ACES clock science. In the past years, the focus has been more on the first two. We have completed the ground terminal site preparation and are ready to receive the ESA ground terminal to be installed at JPL. The ground clock upgrading effort has since shifted to the highly leveraged development of two mercury ion clocks of the next generation for the JPL frequency Standard Test Lab. As the launch date of the ACES payload approaches, more science investigation effort has commenced. In this reporting period, one of the new ground mercury ion clock has been under test and produced encouraging long-term stability data. The second new ground mercury ion clock construction has been completed and now in the operation optimization stage. For the science investigation, we have studied the ACES measurement configurations and analyzed the relativistic observables. The result has been published in a scientific journal. In addition, administrative and logistics of the ground terminal installation have been mostly addressed.
Bibliography Type:	Description: (Last Updated: 10/28/2020)
Articles in Peer-reviewed Journals	Turyshev SG, Yu N, Toth VT. "General relativistic observables for the ACES experiment." Physical Review D. 2016 Feb 15;93(4):045027. https://doi.org/10.1103/PhysRevD.93.045027 , Feb-2016